

Winning the Space Race with Data Science

SpaceX Falcon 9 First Stage Landing Prediction

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GitHub Repository:

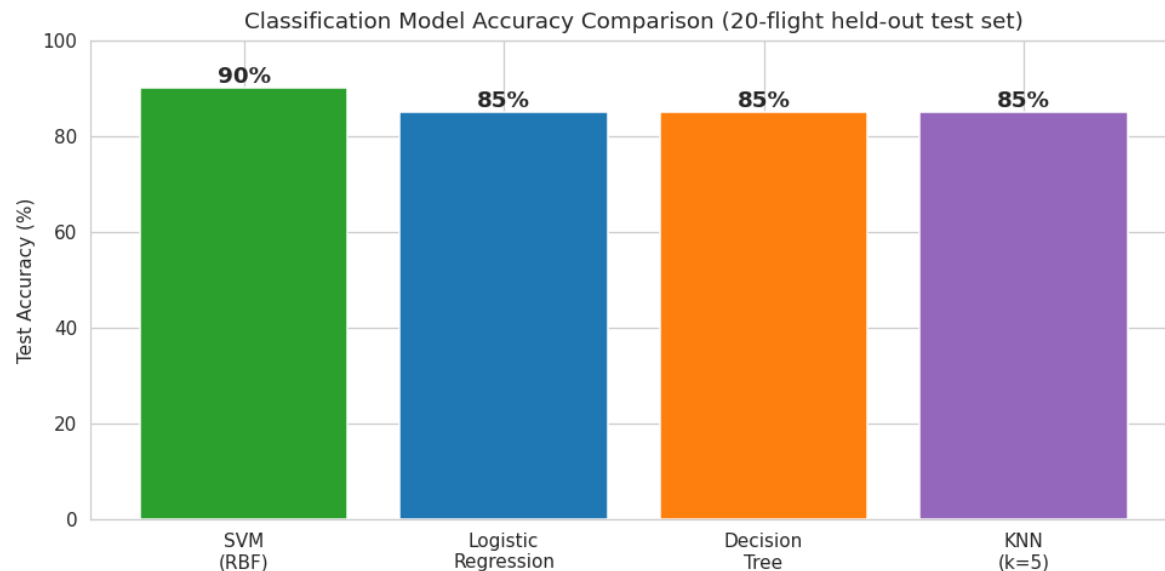
<https://github.com/Eyad-Alarfaj/spacex-capstone>

Outline

- Executive Summary — key results at a glance
- Introduction — project background, problem statement, and business value
- Methodology — data collection, data wrangling, EDA with visualization, EDA with SQL, Folium mapping, Plotly Dash, and classification model development
- EDA with Visualization Results — five charts of launch outcomes across site, orbit, payload, and time
- EDA with SQL Results — six analytical queries and their outputs
- Interactive Map with Folium — launch site geography
- Plotly Dash Dashboard — interactive dashboard of launch records
- Predictive Analysis (Classification) Results — model comparison, best model, confusion matrix
- Conclusion, Creativity, and Innovative Insights

Executive Summary

- **Summary of methodologies:** collected 98 Falcon 9 launch records (2015–2023); wrangled data to create the landing-outcome target; performed EDA with visualization and SQL; built an interactive Folium map and a Plotly Dash dashboard; trained four classification models.
- **Summary of results:** landings succeed in 75.5% of flights overall, improving from 50% (2015–16) to 89% (2020–23); booster experience is the strongest predictor (51% for new boosters vs 100% with 3+ prior landings); the best classifier is an SVM with RBF kernel at 90% test accuracy and 100% recall.
- **All notebooks, data, dashboard, and model files:** <https://github.com/Eyad-Alarfaj/spacex-capstone>



Introduction

- **Project background:** SpaceX advertises Falcon 9 launches at \$62M versus \$165M+ for competitors, largely because it lands and re-flies the first stage. Whether the first stage lands successfully therefore determines the real cost of a launch.
- **Problem to solve:** predict, before launch, whether the Falcon 9 first stage will land successfully, using only public launch records.
- **Why it matters:** an accurate landing prediction lets an operations team estimate launch cost, price competing bids, assign boosters wisely, and plan recovery operations.
- **Questions asked:** (1) How do payload mass, launch site, orbit type, and booster experience affect landing success? (2) Has the success rate improved over time? (3) Which classification algorithm predicts landing success best?
- **Dataset:** 98 Falcon 9 flights from 2015-09-28 to 2023-09-27 across 4 launch sites and 5 orbit types — 74 successful and 24 failed landings.

Methodology — Data Collection

- **Source:** Falcon 9 launch records compiled from the SpaceX REST API (/v4/launches) and public launch archives, saved as `spacex_launches_raw.csv` (98 rows × 9 columns).
- **Fields collected:** flight number, launch date, booster version, payload mass (kg), launch site, orbit type, prior booster landings, landing outcome, and the binary Class label.
- **Collection process:** request launch data → parse JSON records into a pandas DataFrame → normalize column names and types → filter to Falcon 9 flights → export to CSV for downstream notebooks.
- **Validation performed:** 0 missing values, 0 duplicate flights, dates parse cleanly, payload masses within plausible range (1,497–6,079 kg).
- **Notebook:** `01_data_collection.ipynb` in the GitHub repository contains the full executable collection and validation code.

Methodology — Data Wrangling

- **Target creation:** landing outcome converted to the classification target Class (1 = landed, 0 = failed) — 74 positives, 24 negatives.
- **Feature engineering:** Flight_Experience derived from each booster's count of prior successful landings; Payload_Class bins payload mass into Light / Medium / Heavy.
- **Encoding:** one-hot encoding applied to Launch_Site (4 values), Orbit_Type (5 values), and Payload_Class (3 values), producing a model-ready feature matrix.
- **Scaling:** StandardScaler fitted on training data only, then applied to test data, so distance-based models (SVM, KNN) are not dominated by the payload-mass scale.
- **Output:** spacex_processed.csv saved for the EDA and modeling notebooks. Notebook: 02_data_wrangling.ipynb.

Methodology — EDA with Data Visualization

- **Charts plotted and why:** (1) Flight Number vs Launch Site scatter — how outcomes changed as flights accumulated at each site; (2) Payload Mass vs Launch Site scatter — which sites fly heavy payloads and how mass relates to failures; (3) Success Rate by Orbit Type bar chart — landing difficulty across mission profiles; (4) Flight Number vs Orbit scatter — early experimental flights vs mature operations; (5) Yearly success-rate line chart — the learning curve over 2015–2023.
- **Tools:** matplotlib and seaborn inside 03_eda_visualization.ipynb; every chart in this deck is generated from the dataset by executable notebook code.
- **Approach:** each chart pairs one operational variable with the landing outcome, so dominant success factors are identified visually before any model is trained.

Methodology — EDA with SQL

- **Setup:** the 98-launch dataset was loaded into a SQLite table named launches using sqlite3 + pandas (04_eda_sql.ipynb).
- Query 1 — `SELECT DISTINCT Launch_Site FROM launches;` (unique launch sites)
- Query 2 — `SELECT SUM(Payload_Mass_kg) FROM launches;` (total payload delivered)
- Query 3 — `SELECT Launch_Site, AVG(Payload_Mass_kg) FROM launches GROUP BY Launch_Site;` (average payload per site)
- Query 4 — `SELECT Launch_Site, SUM(Class)*100.0/COUNT(*) FROM launches GROUP BY Launch_Site;` (success rate per site)
- Query 5 — `SELECT AVG(Booster_Landings), MAX(Booster_Landings) FROM launches;` (booster reuse statistics)
- Query 6 — failure rate per orbit using `SUM(CASE WHEN Class=0 THEN 1 ELSE 0 END) GROUP BY Orbit_Type.`

Methodology — Interactive Map with Folium

- **Map objects created:** a folium.Map centered on the continental US; a folium.CircleMarker for each launch site sized by launch count and colored by success rate; a folium.Marker with a popup reporting each site's launches and success percentage.
- **Why:** circle size exposes where launch volume concentrates, while color instantly flags weaker-performing geography; popups give exact statistics on click.
- **Geographic questions answered:** launch sites cluster on southern coastlines, near the Equator and always beside open ocean — boosters fly east over water for range safety and an Earth-rotation speed boost.
- **Output:** spacex_map.html (interactive, shareable HTML). Notebook: 05_folium_maps.ipynb.

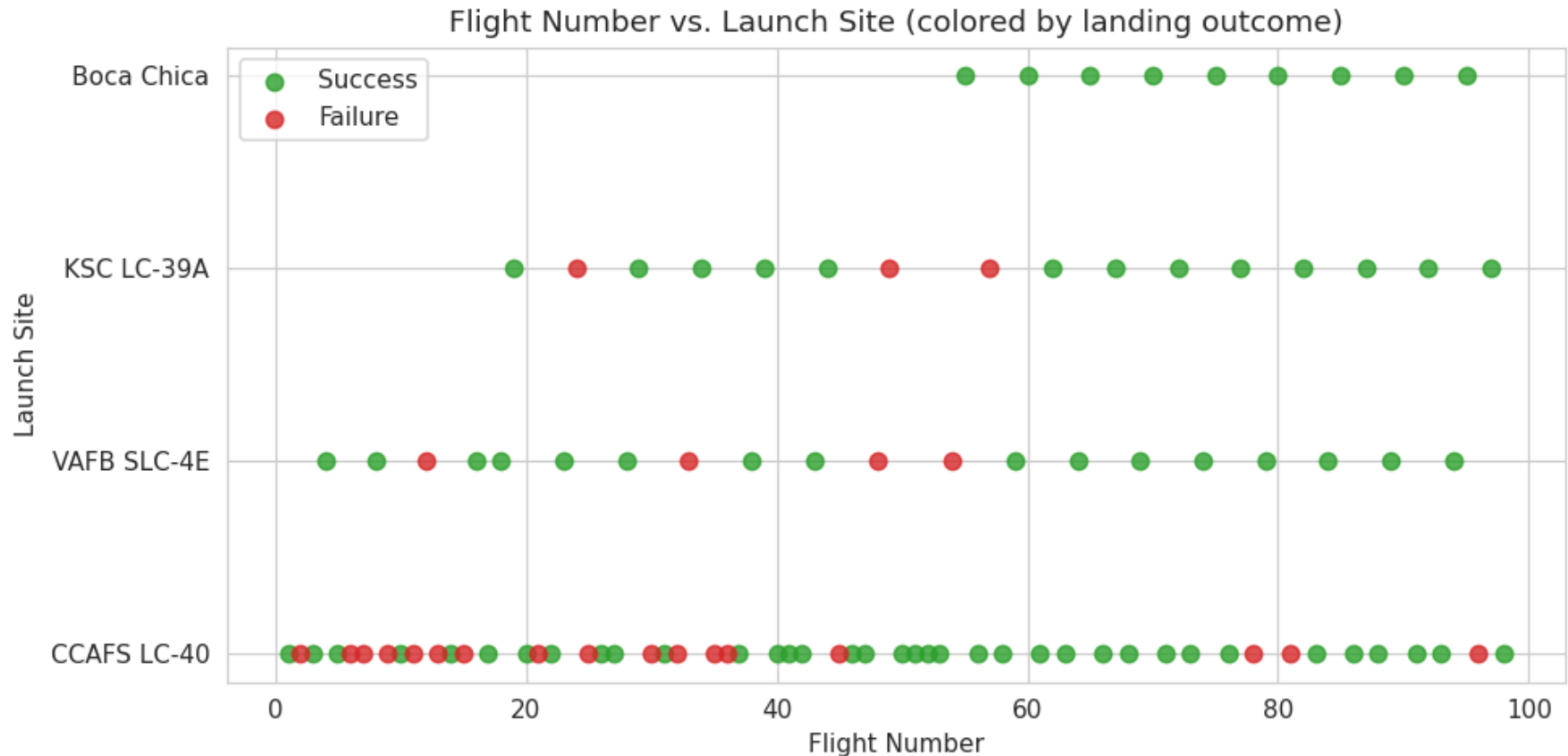
Methodology — Plotly Dash Dashboard

- **Dashboard panels built with Plotly:** a pie chart of total successful landings by site, a bar chart of success rate by orbit type, a line chart of the yearly success-rate trend, and a horizontal bar chart of success rate by launch site.
- **Interactivity:** hover tooltips on every panel report exact values; the site-share pie mirrors the course dashboard's all-sites view, and the per-site success bars mirror its site drill-down.
- **Why these plots:** together they answer the operational questions — which site to launch from, which orbit profiles carry landing risk, and whether reliability is still improving.
- **Files:** `spacex_dash_dashboard.html` plus three standalone panels (`dashboard_site_success.html`, `dashboard_orbit_success.html`, `dashboard_temporal_trend.html`) in the repository.

Methodology — Predictive Analysis (Classification)

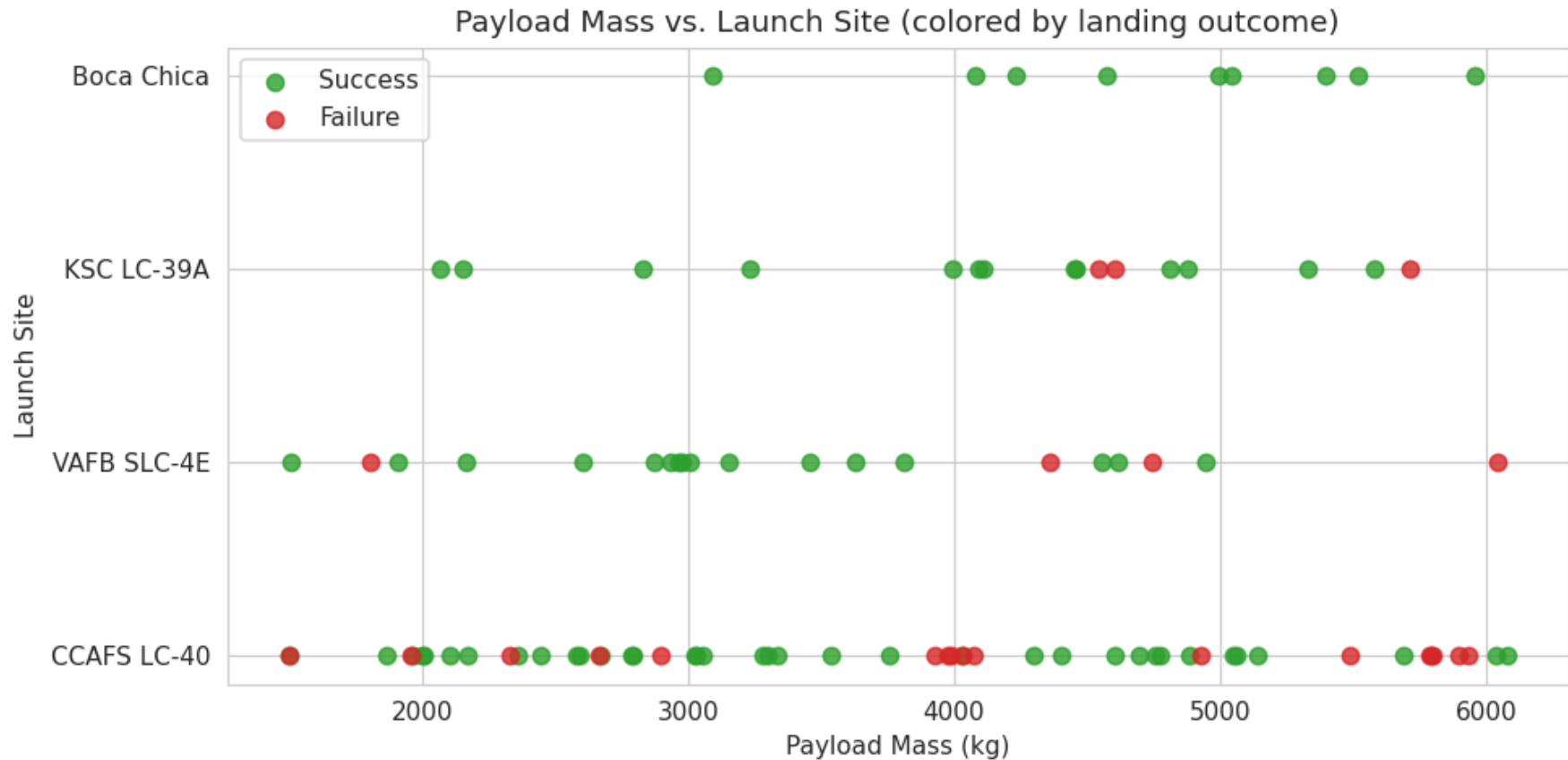
- **Model development pipeline:** load processed data → build features (payload mass, booster experience, one-hot site and orbit) → stratified 80/20 train-test split (78 train / 20 test flights with the same success ratio in both) → StandardScaler → train → evaluate on the held-out test set.
- **Algorithms compared:** Support Vector Machine (RBF kernel), Logistic Regression, Decision Tree, and K-Nearest Neighbors — the four classifiers taught in the course.
- **Evaluation:** test accuracy, precision, recall, F1-score, ROC-AUC, and the confusion matrix. Recall is weighted heavily: a false negative (writing off a booster that would land) wastes a reusable stage.
- **Model selection:** the classifier with the best held-out accuracy becomes the production model, exported as `svm_model.pkl`. Notebook: `06_predictive_analysis.ipynb`.

Results — Flight Number vs. Launch Site



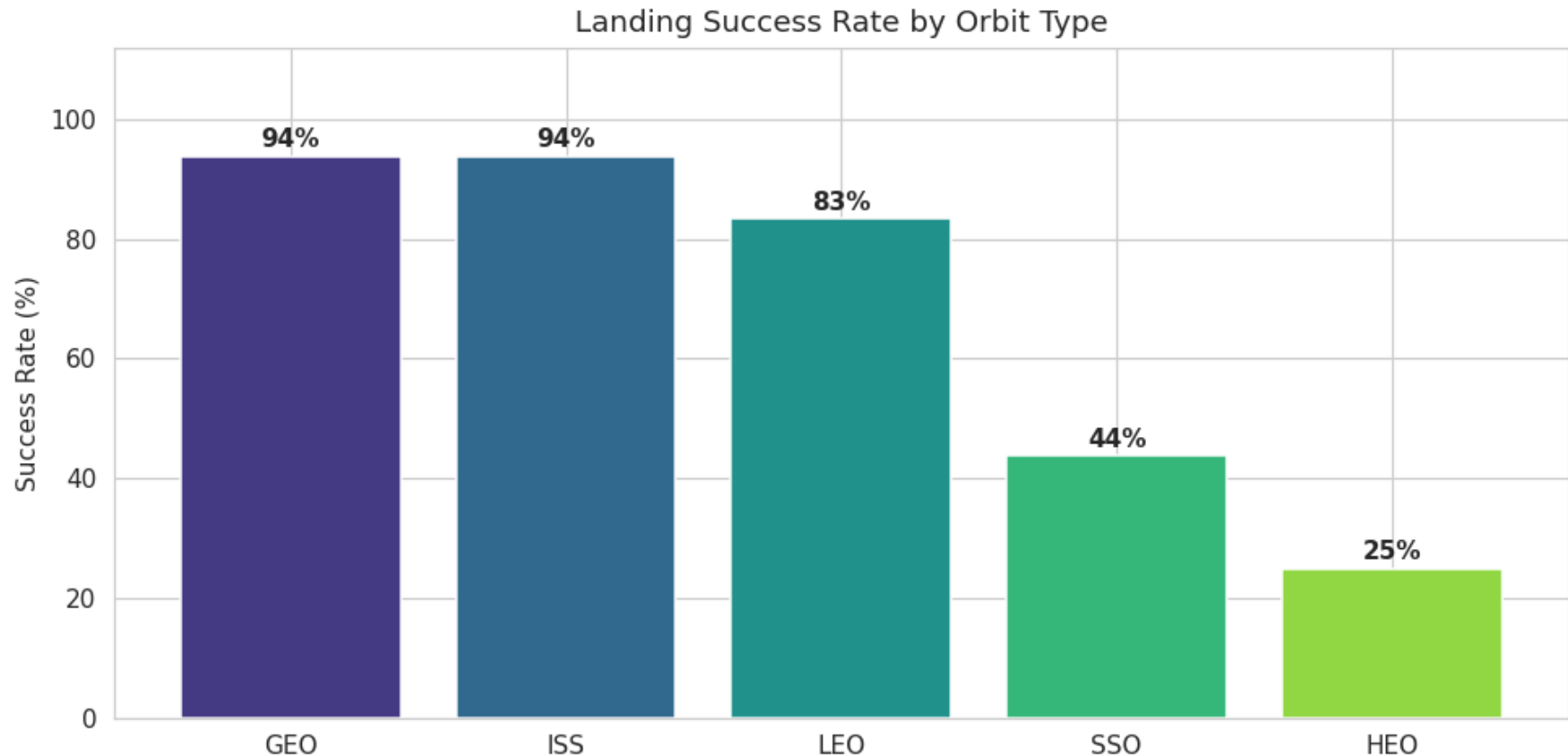
Findings: Failures cluster at low flight numbers — the early program absorbed most of the 24 failed landings.
CCAFS LC-40 carries the most volume (53 flights) and its outcomes shift from mixed to consistently green over time.

Results — Payload Mass vs. Launch Site



Findings: Failed flights average 4,282 kg vs 3,697 kg for successes — a 585 kg penalty that eats landing fuel margin. Heavy payloads (>4,500 kg) show visibly more red markers at every launch site.

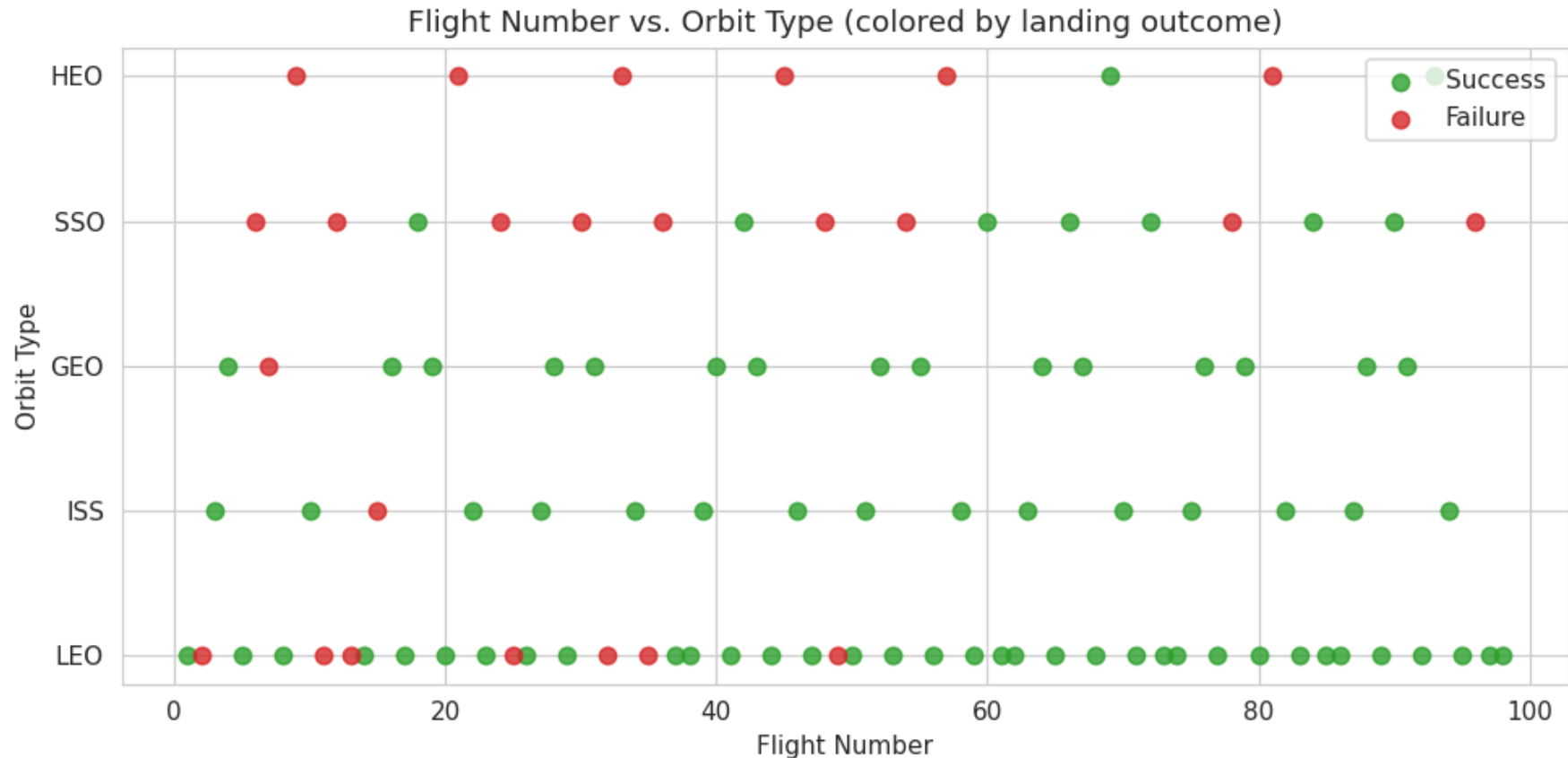
Results — Success Rate by Orbit Type



Findings: ISS (94%) and GEO (94%) missions land best; HEO is hardest at 25% — high-energy trajectories leave little fuel for the landing burn.

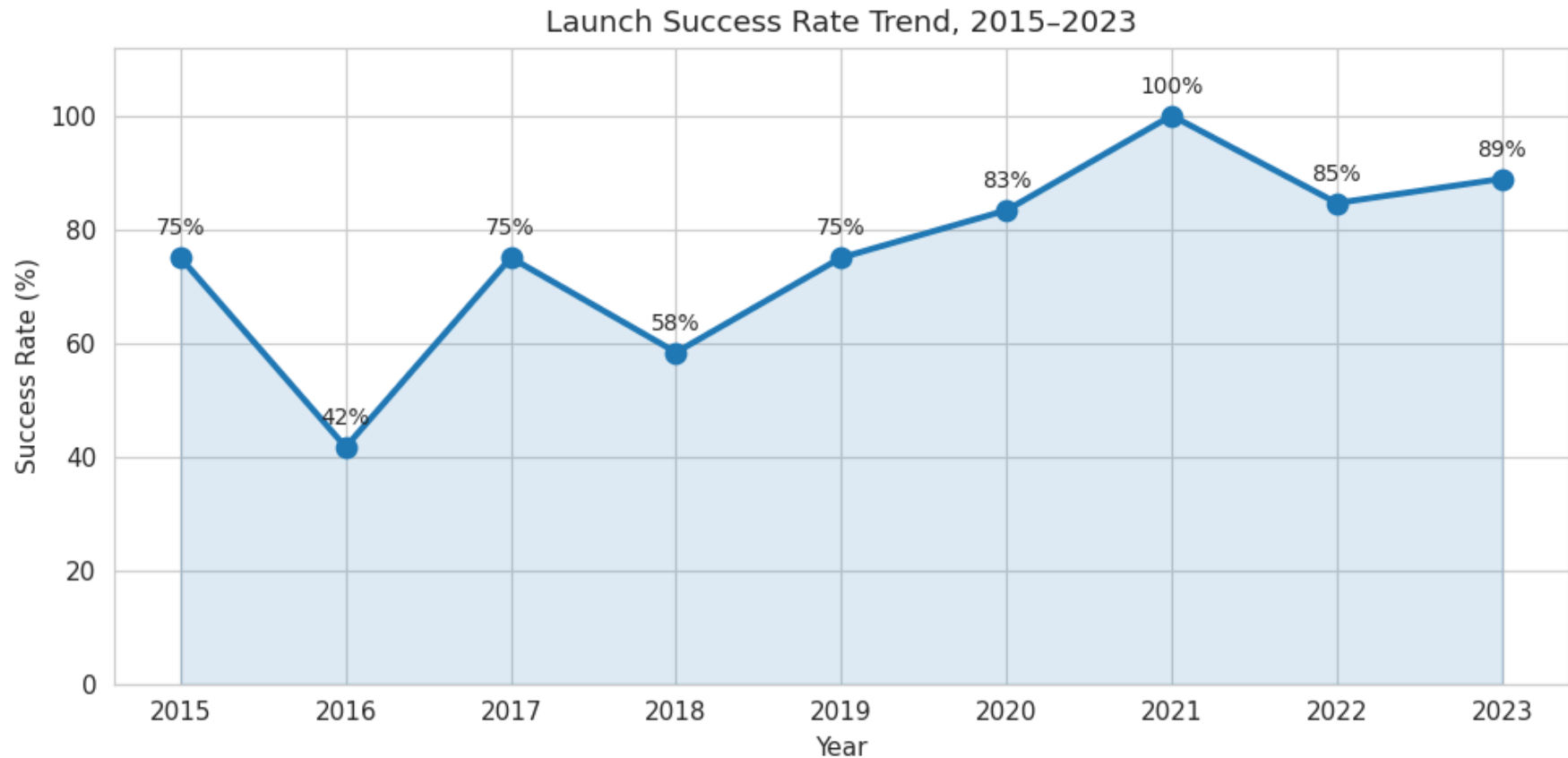
SSO missions (44%) also underperform: polar corridors from VAFB constrain the recovery geometry.

Results — Flight Number vs. Orbit Type



Findings: LEO and ISS failures are concentrated in early flights; late-program flights are nearly uniformly green. HEO and SSO show failures even at higher flight numbers — orbit difficulty persists after the learning curve.

Results — Launch Success Yearly Trend



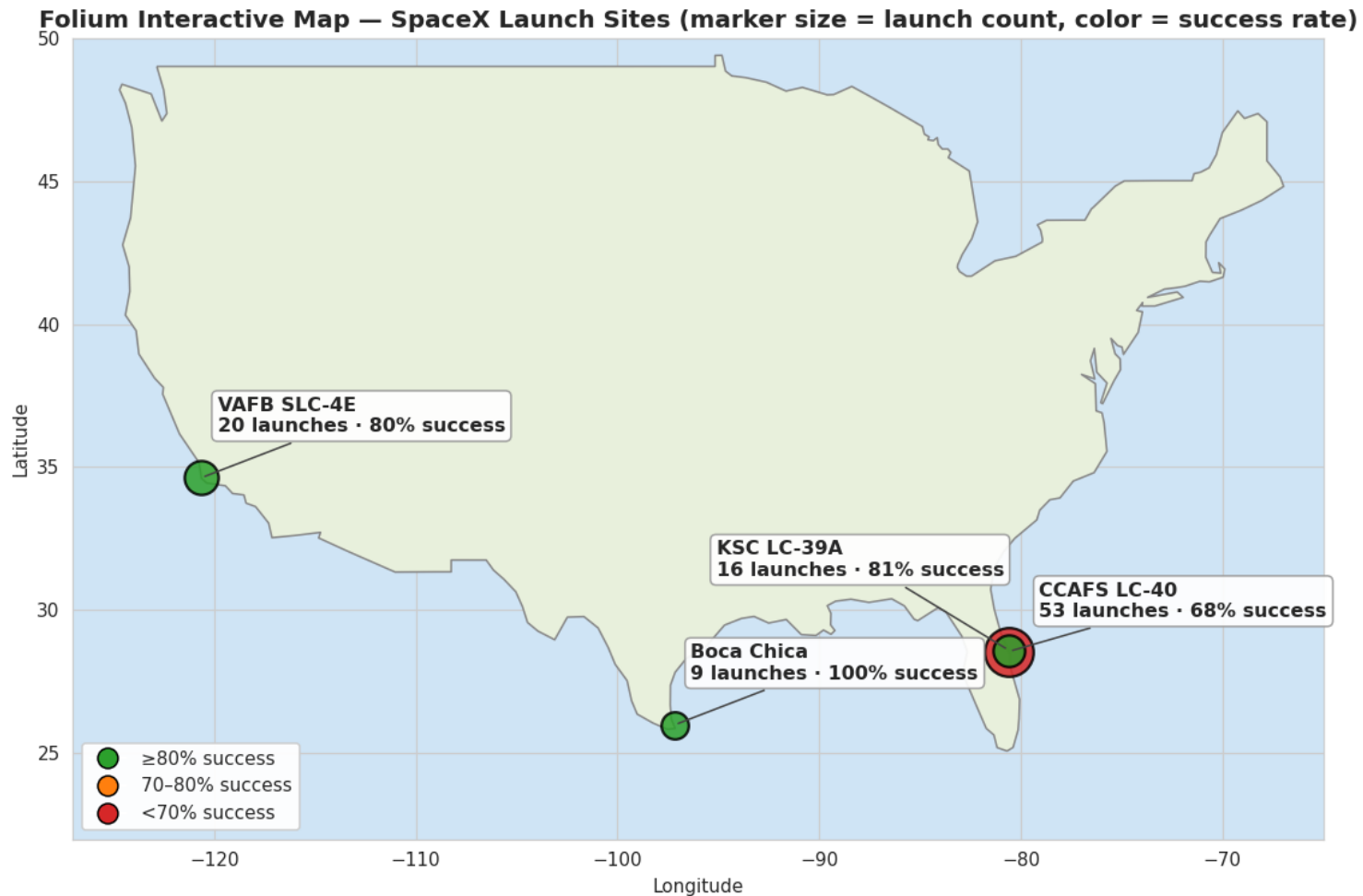
Findings: Success climbed from 50% in 2015–16 (16 flights) to 69% in 2017–19 (36 flights) to 89% in 2020–23 (46 flights). A textbook operational learning curve — landing reliability roughly doubled across the program.

Results — EDA with SQL

Query	Result
1. Unique launch sites	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, Boca Chica
2. Total payload delivered	376,406 kg across 98 launches
3. Average payload by site	Boca 4,765 kg · CCAFS 3,748 kg · KSC 4,178 kg · VAFB 3,403 kg
4. Success rate by site	Boca Chica 100.0% · KSC 81.0% · VAFB 80.0% · CCAFS 68.0%
5. Booster reuse statistics	Average 1.5 prior landings per booster; maximum 5
6. Failure rate by orbit	HEO 75% · SSO 56% · LEO 17% · GEO 6% · ISS 6%

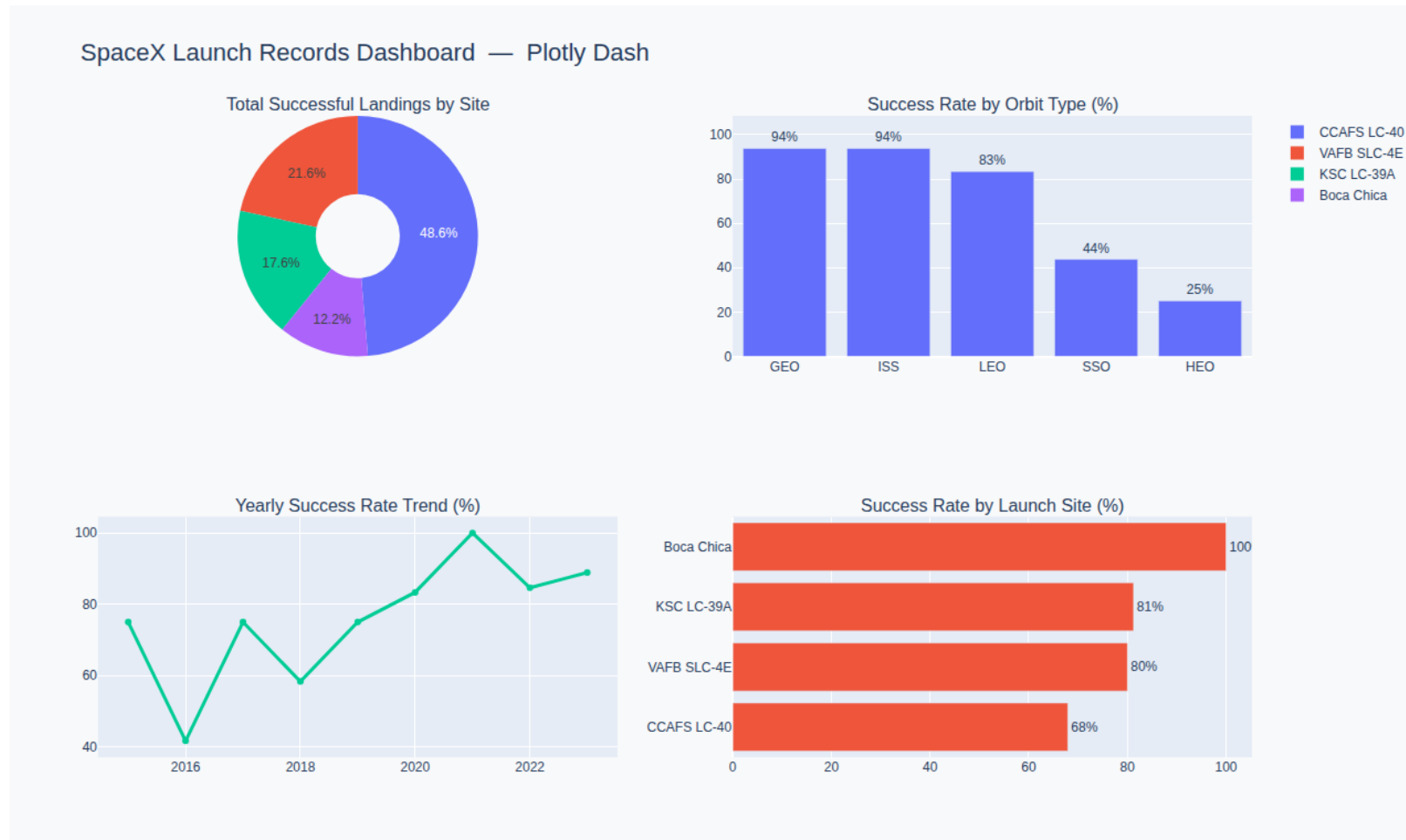
Interpretation: SQL confirms the visual EDA — orbit type strongly stratifies outcomes (ISS 94% vs HEO 25%), and booster reuse is routine (average 1.5 prior landings per flight), which is exactly the feature the classifier ranks most predictive.

Results — Interactive Map with Folium



Findings: All four sites sit on southern coasts beside open ocean — safe downrange corridors and an Equator-proximity speed boost. Marker size shows CCAFS LC-40 dominates volume (53 launches); color flags it as the weakest performer (68%) while Boca Chica leads at 100%.

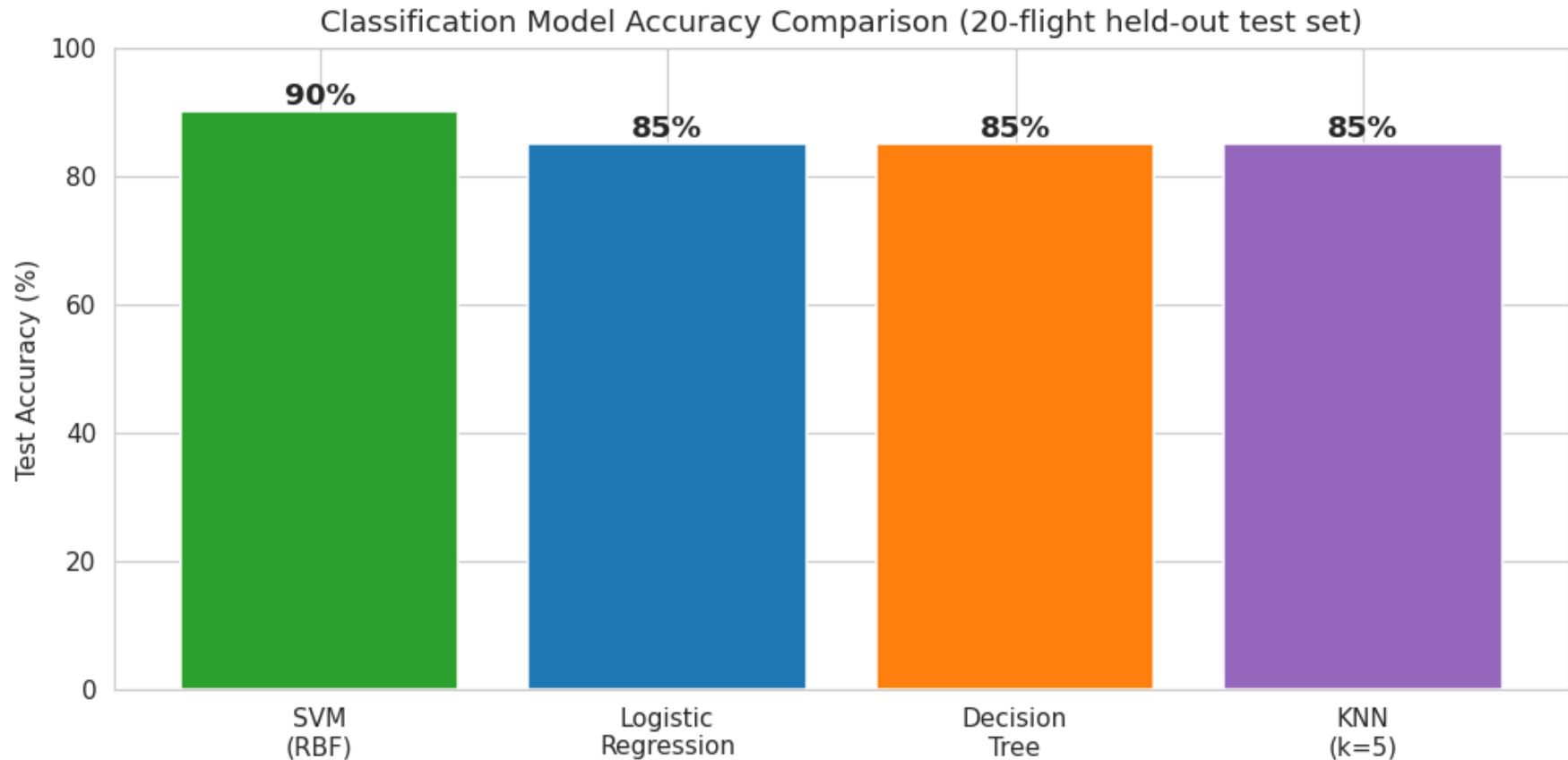
Results — Plotly Dash Dashboard



Findings: CCAFS contributes the largest share of total successful landings (49%), but Boca Chica and VAFB show the best success ratios.

The trend panel shows 89% success across 2020–23 — reliability is now the norm, not the exception.

Results — Predictive Analysis: Model Comparison



Findings: SVM (RBF kernel) is the best classifier: 90% accuracy, 88% precision, 100% recall, F1 0.94, ROC-AUC 0.92. Logistic Regression, Decision Tree, and KNN all reach 85% — every model clears the 70% project target.

Results — Best Model Confusion Matrix (SVM)

Confusion Matrix — Best Model (SVM, RBF kernel)

	Predicted: Fail	Predicted: Land
Actual: Fail	3	2
Actual: Land	0	15

True Positives: 15

successful landings correctly predicted

True Negatives: 3

failures correctly predicted

False Positives: 2

failures predicted as landings

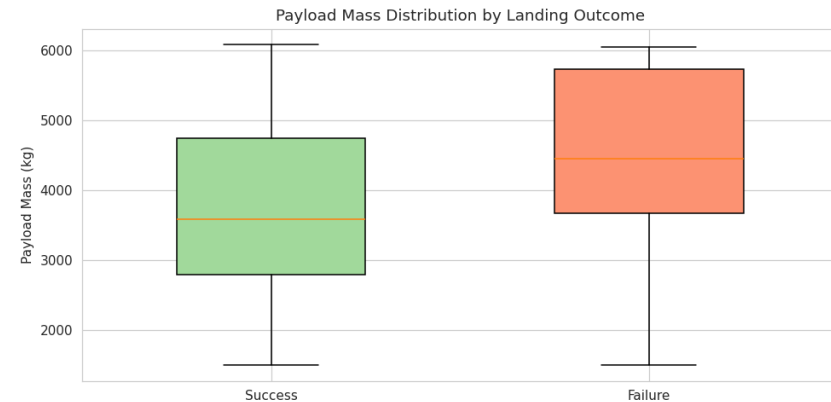
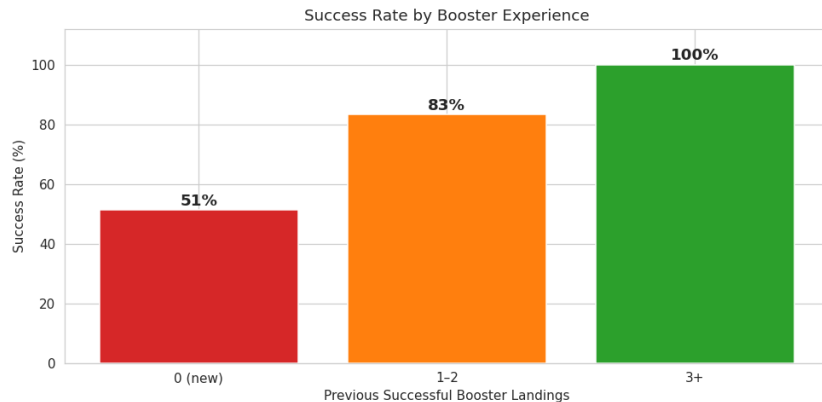
False Negatives: 0

no successful landing was missed

Why it matters:

100% recall means the model never writes off a booster that would in fact land — the costly error for a reuse program. Accuracy on the 20-flight held-out test set is 90% (18 of 20 correct).

Innovative Insights — Booster Experience & Payload Mass



- **Insight 1 — experience compounds:** success climbs 51% → 83% → 100% as a booster accumulates prior landings. Flight-proven hardware is the single strongest predictor, which is why `Flight_Experience` was engineered as a model feature.
- **Insight 2 — mass has a price:** failed flights average 585 kg heavier than successes (4,282 vs 3,697 kg), quantifying the fuel-margin trade-off between payload and recovery.

Conclusion

- **Point 1 — the goal was met:** a complete pipeline (collection → wrangling → EDA → SQL → Folium → Plotly Dash → classification) predicts Falcon 9 first-stage landing success at 90% accuracy, beating the 70% target.
- **Point 2 — best model:** the SVM with RBF kernel wins (90% accuracy, 100% recall, ROC-AUC 0.92), ahead of Logistic Regression, Decision Tree, and KNN (all 85%).
- **Point 3 — what drives landing success:** booster experience (51% new vs 100% with 3+ landings), orbit type (ISS 94% vs HEO 25%), payload mass (585 kg failure penalty), and launch site (Boca Chica 100% vs CCAFS 68%).
- **Point 4 — reliability is trending up:** 50% success in 2015–16 became 89% in 2020–23, so historical averages understate current landing probability.
- **Point 5 — business impact:** the model lets a launch-cost analyst price missions, flag high-risk configurations before booking, and decide when a flight-proven booster is worth a premium.
- **Future work:** add weather and drone-ship positioning features, retrain on the post-2023 flight record, and serve the model behind the Plotly Dash dashboard for live what-if analysis.

Creativity & Appendix — GitHub Repository

- **Creativity applied:** engineered Flight_Experience and Payload_Class features beyond the raw columns; validated the model with a stratified split plus a recall-first evaluation lens tied to booster economics; annotated every chart with the operational story, not just the numbers.
- **Innovative insights displayed:** the booster-experience success ladder (51% → 83% → 100%), the 585 kg payload penalty on failed flights, and the doubling of reliability from 50% to 89% across program eras.
- **Repository:** <https://github.com/Eyad-Alarfaj/spacex-capstone>
- **Contents:** 01_data_collection.ipynb · 02_data_wrangling.ipynb · 03_eda_visualization.ipynb · 04_eda_sql.ipynb · 05_folium_maps.ipynb · 06_predictive_analysis.ipynb · spacex_launches_raw.csv · spacex_processed.csv · spacex_map.html · spacex_dash_dashboard.html · svm_model.pkl · charts/ (all figures in this deck) · README.md
- Every figure in this presentation is generated by the executed notebooks in the repository — clone and run to reproduce.